

A Somos-Type Constraint on Hankel Determinants and Jacobi Recurrence Coefficients

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Abstract

Assuming that a sequence arises as the shifted Hankel transform of the coefficient sequence of a Stieltjes continued fraction, we translate a Somos-type quartic relation into a constraint on the recurrence coefficients of the associated monic orthogonal polynomials. We prove a characterization: the shifted Hankel determinants satisfy the Somos-type relation if and only if consecutive off-diagonal Jacobi coefficients satisfy a fixed symmetric biquadratic equation. Equivalently, the Somos constraint imposes an algebraic law on successive ratios of squared norms. The diagonal Jacobi coefficients do not occur in this condition. We also record an explicit quadratic algebraic generating-function ansatz whose coefficient Hankel determinants reproduce, in the accompanying symbolic computation, the Somos-4 family $A(1, t)$ with prescribed initial determinants.

Preface and motivation

This note originated from a rather specific question concerning Somos-4 sequences and, in particular, from an exchange with Michael Somos about computations involving rational Somos-4 sequences.

The immediate starting point was the observation that the general Somos-4 recurrence is a rational recurrence. Given arbitrary nonzero initial values, there is no reason for the resulting terms to remain integral; rational values are in fact the natural generic situation. The remarkable phenomenon is rather the opposite one: for certain special integer initial conditions, the recurrence produces integers indefinitely. This distinction led to a simple question: what structure remains visible if one deliberately does not restrict attention to the exceptional integral solutions?

The question arose in connection with an experiment using the two elementary transformations which replace one member of a four-term Somos-4 seed by the other root of the corresponding quadratic relation. Starting from four consecutive terms of the classical Somos-4 sequence

$$1, 1, 2, 3,$$

with parameters

$$p_1 = p_2 = 1, \quad p_4 = 4,$$

these transformations generate alternative four-term seeds having the same Somos parameters. Iterating the transformations therefore provides a convenient way of exploring different rational orbits belonging to the same underlying Somos-4 family.

In the course of this experiment, an unexpected connection appeared with OEIS A377264, a sequence arising from a nonlinear recurrence for coefficients of a Stieltjes continued fraction.

This suggested looking at the problem from a direction somewhat different from the usual treatments of Somos sequences.

There is already a rich theory surrounding Somos–4 recurrences, including the Laurent phenomenon, elliptic curves and elliptic divisibility sequences, theta functions, and combinatorial interpretations; see, for example, Fomin and Zelevinsky [5] and Hone and Swart [6]. The purpose here is not to reproduce or compete with those developments. Instead, we deliberately take a different route. The central object is the Somos biquadratic relation

$$a_1^2 a_4^2 + p_1 a_1 a_3^3 + p_1 a_2^3 a_4 + p_2 a_2^2 a_3^2 - p_4 a_1 a_2 a_3 a_4 = 0,$$

viewed not primarily as an elliptic or combinatorial object, but as a relation which may be transported through continued fractions, Hankel determinants, moment functionals, and orthogonal polynomials.

This viewpoint leads naturally to the ratios

$$u_n = \frac{a_{n-1} a_{n+1}}{a_n^2}.$$

For shifted Hankel determinants these ratios have an independent interpretation: they are precisely the off-diagonal coefficients of the associated monic Jacobi recurrence. The Somos biquadratic relation therefore becomes a direct algebraic constraint on consecutive Jacobi coefficients. This observation is the main starting point of the present note.

The connection with continued fractions is particularly useful because it also explains why nonlinear recurrences for individual Stieltjes coefficients can produce Somos sequences after contraction. The example associated with A377264 provides a concrete instance in which the continued-fraction coefficients, their contracted Jacobi coefficients, the Hankel determinants, and the resulting elliptic dynamics can all be seen within the same construction.

The development that follows should therefore be read partly as an exploration. Some of the results are elementary characterizations, while later sections investigate the integrable structure suggested by them. In particular, the resulting two-dimensional map has a symmetric biquadratic invariant, is symplectic and reversible, and possesses genus-one invariant curves. This naturally raises the question of whether the continued-fraction level admits a lift to a Volterra- or modified-Volterra-type transformation. That latter direction is left explicitly as a conjectural one.

The present version also has something of the character of a research notebook. It grew out of the attempt to understand several connections which were not apparent at the beginning, and some of the organization reflects that history. Nevertheless, the individual statements and calculations are presented independently of that correspondence, with the intention that the mathematical content should stand on its own.

The motivating correspondence also served as a useful reminder that there is value in examining a familiar structure outside its most celebrated special cases. In the present setting, the rational solutions are not a nuisance to be removed in favor of the integral ones; they provide the generic setting in which the underlying algebraic and dynamical structure becomes visible. This note is an attempt to follow that structure.

1 Introduction

The purpose of this note is to identify the orthogonal-polynomial content of a Somos-type quartic relation satisfied by shifted Hankel determinants. The main result is a characterization in terms of the off-diagonal coefficients in the three-term recurrence for the monic orthogonal polynomials of the normalized shifted moment functional. More precisely, if these coefficients are denoted by $\hat{b}_n$, then the quartic recurrence for the determinants is equivalent to the biquadratic

relation

$$\widehat{b}_n^2 \widehat{b}_{n+1}^2 - p_4 \widehat{b}_n \widehat{b}_{n+1} + p_1 (\widehat{b}_n + \widehat{b}_{n+1}) + p_2 = 0$$

for all relevant indices. No diagonal coefficient $\widehat{d}_n$ appears. Since $\widehat{b}_n$ is also the ratio of two consecutive squared norms, the same equation gives an algebraic law for successive norm ratios. Liu, Wang, and Zhang [11] study sufficient conditions under which Hankel transforms of generating functions satisfy (α, β) Somos-4 identities. They also use the standard quotient variables

$$x_n = \frac{S_{n-1} S_{n+1}}{S_n^2}$$

and their rational first integral in the analysis of even and odd subsequences. The purpose of the present note is different. Assuming that the terms are first-shifted Hankel determinants of a Stieltjes moment sequence, we identify these quotient variables with the off-diagonal Jacobi coefficients of the normalized shifted moment functional. This yields an if-and-only-if characterization of the Somos identity as a symmetric biquadratic constraint on consecutive Jacobi coefficients. In particular, the diagonal Jacobi coefficients are absent from, and remain unrestricted by, the Somos constraint.

2 The quartic relation

Let $(a_n)_{n \geq 1}$ satisfy

$$a_{n-3}^2 a_n^2 + p_1 a_{n-3} a_{n-1}^3 + p_1 a_{n-2}^3 a_n + p_2 a_{n-2}^2 a_{n-1}^2 - p_4 a_{n-3} a_{n-2} a_{n-1} a_n = 0. \quad (1)$$

Here p_1, p_2, p_4 are fixed parameters. The entries may lie in $\mathbb{R}$, $\mathbb{C}$, or a suitable commutative field; no integrality assumption is made.

3 Stieltjes continued fraction and moments

Classical connections among continued fractions, moments, and Hankel determinants are discussed by Wall [2] and Flajolet [4]. Consider the formal Stieltjes continued fraction

$$S(x) = \frac{1}{1 - \frac{c_1 x}{1 - \frac{c_2 x}{1 - \frac{c_3 x}{\ddots}}}} = \sum_{n \geq 0} \mu_n x^n. \quad (2)$$

Its constant coefficient is $\mu_0 = 1$. Thus no additional coefficient c_0 is hidden in (2). If a numerator c_0 is placed in front of the fraction, then the moment sequence is merely rescaled by c_0 . We use the normalization $c_0 = 1$. For $r, m \geq 0$, define

$$\Delta_m^{(r)} = \det(\mu_{i+j+r})_{0 \leq i, j \leq m-1}, \quad \Delta_0^{(r)} = 1. \quad (3)$$

The sequence considered here is the first shifted Hankel sequence

$$a_n = \Delta_n^{(1)} \quad (n \geq 1). \quad (4)$$

For (2),

$$\Delta_n^{(1)} = c_1^n \prod_{j=1}^{n-1} (c_{2j} c_{2j+1})^{n-j}. \quad (5)$$

In particular,

$$a_1 = c_1, \quad a_2 = c_1^2 c_2 c_3, \quad a_3 = c_1^3 c_2^2 c_3^2 c_4 c_5, \quad a_4 = c_1^4 c_2^3 c_3^3 c_4^2 c_5^2 c_6 c_7. \quad (6)$$

4 Continued fraction contraction

The even contraction of (2) is

$$S(x) = \frac{1}{1 - d_0x - \frac{b_1x^2}{1 - d_1x - \frac{b_2x^2}{1 - d_2x - \ddots}}}, \quad (7)$$

where

$$d_0 = c_1, \quad d_n = c_{2n} + c_{2n+1}, \quad b_n = c_{2n-1}c_{2n} \quad (n \geq 1). \quad (8)$$

The associated monic orthogonal polynomials satisfy

$$P_{-1}(t) = 0, \quad P_0(t) = 1, \quad P_{n+1}(t) = (t - d_n)P_n(t) - b_nP_{n-1}(t). \quad (9)$$

5 The normalized shifted functional

The determinants in (4) belong to the shifted moments (μ_{n+1}) . Normalize them by setting

$$\hat{\mu}_n = \frac{\mu_{n+1}}{c_1}, \quad \hat{L}(t^n) = \hat{\mu}_n. \quad (10)$$

Let $(\hat{P}_n)$ be the monic orthogonal polynomials for $\hat{L}$ and write their recurrence as

$$\hat{P}_{-1}(t) = 0, \quad \hat{P}_0(t) = 1, \quad \hat{P}_{n+1}(t) = (t - \hat{d}_n)\hat{P}_n(t) - \hat{b}_n\hat{P}_{n-1}(t). \quad (11)$$

The shifted product formula identifies the off-diagonal coefficients as

$$\hat{b}_n = c_{2n}c_{2n+1} \quad (n \geq 1), \quad (12)$$

and consequently

$$a_n = c_1^n \prod_{j=1}^{n-1} \hat{b}_j^{n-j}. \quad (13)$$

Thus the Jacobi parameters relevant to (a_n) are the $\hat{b}_n$ of the normalized shifted functional, rather than the original $b_n = c_{2n-1}c_{2n}$.

6 The determinant-ratio reduction

Define

$$F(x, y) = x^2y^2 - p_4xy + p_1(x + y) + p_2. \quad (14)$$

Theorem 1. *Let $(a_n)_{n \geq 1}$ be nonzero, and define*

$$q_n = \frac{a_{n-1}a_{n+1}}{a_n^2} \quad (n \geq 2). \quad (15)$$

For each $n \geq 4$, the quartic relation (1) is equivalent to

$$F(q_{n-2}, q_{n-1}) = 0. \quad (16)$$

Proof. Divide (1) by $a_{n-3}a_{n-2}a_{n-1}a_n$. Using

$$q_{n-2} = \frac{a_{n-3}a_{n-1}}{a_{n-2}^2}, \quad q_{n-1} = \frac{a_{n-2}a_n}{a_{n-1}^2},$$

one obtains

$$q_{n-2}q_{n-1} + \frac{p_1}{q_{n-2}} + \frac{p_1}{q_{n-1}} + \frac{p_2}{q_{n-2}q_{n-1}} - p_4 = 0.$$

Multiplication by $q_{n-2}q_{n-1}$ gives (16). Every step is reversible under the non-vanishing assumptions. $\square$

Proposition 2. *For the shifted Hankel determinants (13),*

$$\frac{a_{n-1}a_{n+1}}{a_n^2} = \hat{b}_n = c_{2n}c_{2n+1}. \quad (17)$$

Proof. In the quotient $a_{n-1}a_{n+1}/a_n^2$, the factors c_1 cancel. For every $j < n$, the exponent of $\hat{b}_j$ is

$$(n-1-j) + (n+1-j) - 2(n-j) = 0,$$

while $\hat{b}_n$ occurs once in a_{n+1} . This proves (17). $\square$

7 Characterization by the polynomial recurrence

Theorem 3 (Characterization theorem). *Let $\hat{L}$ be a normalized quasi-definite moment functional, and let its monic orthogonal polynomials satisfy (11), with $\hat{b}_n \neq 0$. Let (a_n) be the corresponding first shifted Hankel determinants, with the normalization in (13). Then the following are equivalent:*

1. (a_n) satisfies the Somos-type relation (1) for every $n \geq 4$;
2. the off-diagonal coefficients in the polynomial recurrence satisfy

$$F(\hat{b}_m, \hat{b}_{m+1}) = 0 \quad (m \geq 2), \quad (18)$$

that is,

$$\hat{b}_m^2 \hat{b}_{m+1}^2 - p_4 \hat{b}_m \hat{b}_{m+1} + p_1(\hat{b}_m + \hat{b}_{m+1}) + p_2 = 0. \quad (19)$$

The diagonal coefficients $\hat{d}_n$ are unrestricted by this equivalence and do not occur in (19).

Proof. By Proposition 2, $q_m = \hat{b}_m$. For $n \geq 4$, Theorem 1 states that (1) is equivalent to $F(q_{n-2}, q_{n-1}) = 0$. Putting $m = n - 2$ gives (18) for $m \geq 2$. Since neither the determinant product (13) nor the resulting equation contains $\hat{d}_n$, no condition on the diagonal coefficients arises. $\square$

Remark 4. The characterization is entirely off-diagonal. Once a nonzero sequence $(\hat{b}_n)$ satisfying (19) has been chosen, the Somos constraint itself places no additional restriction on the diagonal Jacobi parameters $(\hat{d}_n)$. Questions of quasi-definiteness, positivity, support, and recognition of a resulting measure may impose further conditions, but these are separate from the Somos relation.

8 The associated algebraic curve and dynamics

Every pair $(\widehat{b}_m, \widehat{b}_{m+1})$, $m \geq 2$, lies on

$$C : \quad F(x, y) = 0. \quad (20)$$

The equation is symmetric in x and y and defines a reversible algebraic correspondence. Solving for y gives

$$y = \frac{p_4x - p_1 \pm \sqrt{(p_4x - p_1)^2 - 4x^2(p_1x + p_2)}}{2x^2}. \quad (21)$$

Proposition 5. *Assume $p_1 \neq 0$ and that*

$$D(x) = -4p_1x^3 + (p_4^2 - 4p_2)x^2 - 2p_1p_4x + p_1^2 \quad (22)$$

has no multiple root. Then the smooth projective model of C is a genus-one curve.

Proof. Regarding $F(x, y) = 0$ as a quadratic equation in y and introducing $z = 2x^2y + p_1 - p_4x$ gives the birational model $z^2 = D(x)$. If D is a square-free cubic, this is a nonsingular hyperelliptic curve of genus one. $\square$

Given $(x, y) \in C$, the equation $F(y, z) = 0$ has the backtracking root $z = x$. When $y \neq 0$, the other root is

$$z = \frac{p_4y - p_1}{y^2} - x.$$

Selecting this branch defines the birational map

$$T(x, y) = \left(y, \frac{p_4y - p_1}{y^2} - x \right), \quad (23)$$

where it is defined. Direct substitution shows that T preserves F . Without a branch choice, (20) is naturally a reversible algebraic correspondence; with (23), it is a birational discrete dynamical system on C .

9 Squared norms

Let

$$h_n = \widehat{L}(\widehat{P}_n^2) \quad (24)$$

be the squared norms for the normalized shifted functional. The monic three-term recurrence gives

$$h_n = \widehat{b}_n h_{n-1}, \quad \widehat{b}_n = \frac{h_n}{h_{n-1}}. \quad (25)$$

Corollary 6 (Norm-ratio characterization). *Under the hypotheses of Theorem 3, the shifted Hankel determinants satisfy (1) for every $n \geq 4$ if and only if*

$$F\left(\frac{h_m}{h_{m-1}}, \frac{h_{m+1}}{h_m}\right) = 0 \quad (m \geq 2). \quad (26)$$

Equivalently,

$$\left(\frac{h_{m+1}}{h_{m-1}}\right)^2 - p_4 \frac{h_{m+1}}{h_{m-1}} + p_1 \left(\frac{h_m}{h_{m-1}} + \frac{h_{m+1}}{h_m}\right) + p_2 = 0. \quad (27)$$

Proof. Substitute (25) into (19) and use $\widehat{b}_m \widehat{b}_{m+1} = h_{m+1}/h_{m-1}$. $\square$

10 Somos Parameters and the Associated First Integral

We use the notation $A(p_1, p_2)$ for the autonomous Somos–4 recurrence

$$a_{n+4}a_n = p_1a_{n+3}a_{n+1} + p_2a_{n+2}^2. \quad (28)$$

Equivalently, after shifting the index,

$$a_{n-2}a_{n+2} = p_1a_{n-1}a_{n+1} + p_2a_n^2.$$

Thus p_1 and p_2 are the two fixed coefficients of the Somos recurrence. They are parameters of the recurrence itself and are independent of n . The notation $A(p_1, p_2)$ will always refer to the recurrence (28), together with initial values for which the required denominators are nonzero. The classical Somos–4 recurrence is the specialization

$$A(1, 1) : \quad a_{n+4}a_n = a_{n+3}a_{n+1} + a_{n+2}^2.$$

More generally, $A(1, t)$ refers to $a_{n+4}a_n = a_{n+3}a_{n+1} + ta_{n+2}^2$. To pass from the bilinear recurrence to the Jacobi coefficients, define

$$u_n := \frac{a_{n-1}a_{n+1}}{a_n^2}, \quad (29)$$

whenever the quotient is defined. Dividing the shifted form of (28) by a_n^2 gives

$$u_{n-1}u_n^2u_{n+1} = p_1u_n + p_2. \quad (30)$$

Indeed, $u_{n-1}u_n^2u_{n+1} = a_{n-2}a_{n+2}/a_n^2$. For two consecutive values $x = u_n$ and $y = u_{n+1}$, define

$$H(x, y) = xy + p_1 \left(\frac{1}{x} + \frac{1}{y} \right) + \frac{p_2}{xy}. \quad (31)$$

Along every nonsingular orbit of (30), $H(u_n, u_{n+1})$ is independent of n . We denote this constant by p_4 :

$$p_4 = u_nu_{n+1} + p_1 \left(\frac{1}{u_n} + \frac{1}{u_{n+1}} \right) + \frac{p_2}{u_nu_{n+1}}. \quad (32)$$

The notation p_4 is chosen because this constant is associated with the fourth-order bilinear Somos recurrence. It is not a third coefficient in (28): p_1, p_2 are fixed recurrence coefficients, whereas p_4 is determined by p_1, p_2 and the initial data. Multiplying (32) by u_nu_{n+1} yields

$$u_n^2u_{n+1}^2 - p_4u_nu_{n+1} + p_1(u_n + u_{n+1}) + p_2 = 0. \quad (33)$$

Conversely,

$$u_{n+2} = \frac{p_4u_{n+1} - p_1}{u_{n+1}^2} - u_n. \quad (34)$$

Lemma 7. *Let $u_n \neq 0$ satisfy (30). Then $H(u_n, u_{n+1}) = H(u_{n-1}, u_n)$. Consequently, p_4 in (32) is independent of n .*

Proof. From (30), $u_{n-1} = (p_1u_n + p_2)/(u_n^2u_{n+1})$. Substitution into $H(u_{n-1}, u_n)$ followed by simplification gives $H(u_n, u_{n+1})$. $\square$

Remark 8. The subscript in p_4 should not be interpreted as an index of a sequence. It records the connection with Somos–4 and distinguishes the first integral from the recurrence coefficients p_1 and p_2 . Formula (32) is the definitive meaning of p_4 in this paper.

11 The Example Recorded as OEIS A377264

A concrete continued-fraction example related to the construction above is recorded in the OEIS as A377264. The OEIS entry begins with a sequence of continued-fraction coefficients $d(k)$ satisfying

$$d(k) = \frac{d(k-3)d(k-2) + 1}{d(k-5)d(k-4)d(k-3)^2d(k-2)^2d(k-1)}, \quad (35)$$

with $d(0), d(1), d(2), d(3), d(4) = 1, 1, 1, 2, 1$. The sequence A377264 itself is defined by $\text{num}(d(2n+1))$, where num denotes the numerator in lowest terms. The same OEIS entry considers the Stieltjes continued fraction

$$S(x) = \frac{1}{1 - \frac{d(0)x}{1 - \frac{d(1)x}{1 - \frac{d(2)x}{\ddots}}}} = \sum_{m \geq 0} s_m x^m. \quad (36)$$

Its Hankel transform is a shifted copy of the classical Somos-4 sequence A006720. More generally, the construction uses coefficients $c(k)$ satisfying

$$c(2k+1) = \frac{c(2k-2)c(2k-1) + t}{c(2k-4)c(2k-3)c(2k-2)^2c(2k-1)^2c(2k)}, \quad (37)$$

with a prescribed convention for negative subscripts. The associated Hankel determinants satisfy a Somos-4 recurrence of type $A(1, t)$. Three different sequences occur: the continued-fraction coefficients $d(k)$; A377264, formed from the numerators of $d(2n+1)$; and the Hankel determinants of the coefficient sequence of (36). Accordingly, A377264 should not be identified directly with either the Hankel sequence a_n or the Jacobi sequence $\hat{b}_n$. Under contraction,

$$\hat{b}_n = c_{2n}c_{2n+1}, \quad (38)$$

subject to the initial-index convention adopted for the shifted functional. Thus the Jacobi variables are products of adjacent continued-fraction coefficients, not the odd coefficients alone. In the specialization $t = 1$, the Hankel determinants satisfy $A(1, 1)$, while p_4 is determined from two consecutive nonzero Jacobi coefficients by (32).

Remark 9. The OEIS entry also describes A377264 by odd multiples of a point on an elliptic curve. This provides independent evidence that the continued-fraction recurrence, its Hankel transform, and the associated Jacobi dynamics belong to the same elliptic setting. Because the OEIS formula concerns the numerators of $d(2n+1)$, whereas the present Jacobi variables are the products $d(2n)d(2n+1)$, the two elliptic descriptions should be compared through the continued-fraction contraction rather than by direct termwise identification.

11.1 A generalized continued-fraction construction

The recurrence appearing in the OEIS construction admits a direct generalization to arbitrary Somos parameters p_1, p_2 . Let the Stieltjes coefficients c_k satisfy

$$c_{2k+1} = \frac{p_1 c_{2k-2} c_{2k-1} + p_2}{c_{2k-4} c_{2k-3} c_{2k-2}^2 c_{2k-1}^2 c_{2k}}, \quad (39)$$

with suitable prescribed initial values and with all required denominators nonzero. Define

$$u_k = c_{2k} c_{2k+1}. \quad (40)$$

Since $c_{2k-4}c_{2k-3} = u_{k-2}$ and $c_{2k-2}c_{2k-1} = u_{k-1}$, multiplication of the defining recurrence by c_{2k} gives

$$u_k = \frac{p_1 u_{k-1} + p_2}{u_{k-2} u_{k-1}^2}. \quad (41)$$

Equivalently,

$$u_{k-2} u_{k-1}^2 u_k = p_1 u_{k-1} + p_2. \quad (42)$$

Thus, after an index shift,

$$u_{n-1} u_n^2 u_{n+1} = p_1 u_n + p_2, \quad (43)$$

which is precisely the determinant-ratio recurrence associated with

$$a_{n+4} a_n = p_1 a_{n+3} a_{n+1} + p_2 a_{n+2}^2. \quad (44)$$

Consequently, the corresponding shifted Hankel determinants form a sequence of type $A(p_1, p_2)$.

The parameters p_1, p_2 do not determine the particular orbit. The additional quantity p_4 is the first integral

$$p_4 = u_n u_{n+1} + p_1 \left(\frac{1}{u_n} + \frac{1}{u_{n+1}} \right) + \frac{p_2}{u_n u_{n+1}}, \quad (45)$$

and is independent of n . Equivalently, the initial pair (u_0, u_1) must satisfy

$$u_0^2 u_1^2 - p_4 u_0 u_1 + p_1(u_0 + u_1) + p_2 = 0. \quad (46)$$

Hence, if p_1, p_2, p_4 are prescribed, one may choose an admissible nonzero value $u_0 = r$ and determine u_1 from

$$u_1 = \frac{p_4 r - p_1 \pm \sqrt{(p_4 r - p_1)^2 - 4r^2(p_1 r + p_2)}}{2r^2}. \quad (47)$$

The subsequent values are then generated by

$$u_{n+1} = \frac{p_1 u_n + p_2}{u_{n-1} u_n^2}. \quad (48)$$

Thus p_4 is incorporated through the choice of the initial point on the invariant biquadratic curve, rather than as an additional coefficient of the Somos recurrence.

11.2 Example: $p_1 = p_2 = 1$ and $p_4 = 4$

As a concrete example, consider $p_1 = p_2 = 1$ and $p_4 = 4$. The initial values satisfy

$$u_0^2 u_1^2 - 4u_0 u_1 + u_0 + u_1 + 1 = 0. \quad (49)$$

Choosing $u_0 = 1$ gives $u_1^2 - 3u_1 + 2 = 0$, and hence $u_1 = 1$ or $u_1 = 2$. The choice $u_1 = 1$ gives the constant solution $u_n = 1$. The nontrivial choice $u_0 = 1, u_1 = 2$ generates

$$u_{n+1} = \frac{u_n + 1}{u_{n-1} u_n^2}. \quad (50)$$

The first terms are

$$u_0, u_1, u_2, u_3, u_4, \dots = 1, 2, \frac{3}{4}, \frac{14}{9}, \frac{69}{49}, \dots$$

The corresponding shifted Hankel determinants therefore satisfy the classical Somos-4 recurrence

$$a_{n+4} a_n = a_{n+3} a_{n+1} + a_{n+2}^2,$$

that is, they belong to the family $A(1, 1)$.

At the level of Stieltjes coefficients, any factorization $c_{2n}c_{2n+1} = u_n$ gives the same contracted Jacobi coefficients. In particular, one may choose

$$c_{2n} = r_n, \quad c_{2n+1} = \frac{u_n}{r_n},$$

where the nonzero quantities r_n are arbitrary. This freedom does not change the sequence u_n , the associated Hankel determinants, or the value of p_4 . Thus the three levels of data can be distinguished as follows:

$$(p_1, p_2) \longrightarrow \text{Somos family},$$

$$(p_1, p_2, p_4) \longrightarrow \text{invariant elliptic curve},$$

and

$$(p_1, p_2, p_4, u_0) \longrightarrow \text{particular elliptic orbit}.$$

The remaining factorization variables r_n do not affect the contracted Jacobi dynamics.

11.3 An integral-moment gauge for the family $A(1, N)$

The factorization freedom can be used to impose an additional arithmetic condition when integral moments are desired. A concrete construction for the family $A(1, N)$ is recorded in OEIS A377441 [13]. We retain the indexing used in that entry in order to avoid conflating it with the one-based coefficients c_j of (2).

Let $d(m, N)$ be defined, with the initial convention of A377441, by

$$d(m, N) = \begin{cases} \frac{d(m-3, N)d(m-2, N) + N}{d(m-5, N)d(m-4, N)d(m-3, N)^2d(m-2, N)^2d(m-1, N)}, & m \text{ even}, \\ \frac{1}{d(m-1, N)d(m-2, N)}, & m \text{ odd}. \end{cases} \quad (51)$$

The convention for the initial indices is $d(m, N) = 1$ before the recurrence starts, as specified in the OEIS entry. The associated Stieltjes continued fraction produces the integer moment rows recorded in A377441. Its shifted Hankel determinants satisfy the Somos-4 recurrence of type $A(1, N)$.

The structurally important additional condition is the reciprocal parity rule

$$d(m, N)d(m-1, N)d(m-2, N) = 1 \quad (m \text{ odd}). \quad (52)$$

This condition is not imposed by the Somos recurrence. Rather, it fixes the otherwise free factorization at every second step. In the notation of the present paper, the contracted quantities remain products of adjacent Stieltjes coefficients, whereas the reciprocal parity rule selects particular individual factors above that contracted orbit.

To see the distinction, suppose that a contracted coefficient is written as

$$u_n = c_{2n}c_{2n+1}.$$

Without an additional condition, one may choose

$$c_{2n} = r_n, \quad c_{2n+1} = \frac{u_n}{r_n}$$

with arbitrary nonzero r_n . An A377441-type reciprocal relation removes this arbitrariness recursively. The projected Jacobi dynamics and the shifted Hankel determinants are preserved, but the selected moment sequence is changed. For $A(1, N)$, the construction in A377441 selects a representative having integral moments even though its individual continued-fraction coefficients are generally rational.

Thus moment integrality is not a consequence of the Somos relation alone, nor of integrality of the individual Stieltjes coefficients. It is an additional property of the chosen lift from the contracted Jacobi coefficients to the continued-fraction coefficients. For the family $A(1, N)$, the recurrence above supplies an explicit optional constraint that enforces this arithmetic choice.

Remark 10. The indexing of A377441 and the indexing of the shifted determinants in the present paper differ by initial terms. Accordingly, the continued-fraction coefficients and the contracted products should be compared only after applying the corresponding shift. The arithmetic statement is unaffected: the A377441 construction gives an integral moment sequence whose Hankel transform belongs to the family $A(1, N)$.

12 Direct generation from an algebraic generating function

The preceding constructions begin with continued-fraction or Jacobi data and then form Hankel determinants. Here we show how to generate the Somos-4 sequence directly from the generating function of moments.

Let λ, η, τ be parameters and let

$$G(z) = \sum_{n \geq 0} g_n z^n.$$

Define the three polynomials

$$A(z) = \lambda(\eta\tau - \eta^2)z^2 + (-(\lambda^2\eta + 2)\tau + \lambda^2\eta^2 + 2\eta - 2\lambda)z + \lambda(\tau + \lambda - \eta), \quad (53)$$

$$B(z) = -\lambda^2\eta\tau z^2 + ((2\lambda^3\eta + 3\lambda)\tau - \lambda^3\eta^2 - \lambda\eta + 3\lambda^2)z + \lambda^2(\eta - 2\lambda - 2\tau), \quad (54)$$

$$C(z) = -((\lambda^4\eta + \lambda^2)\tau + \lambda^3)z + \lambda^3(\tau + \lambda). \quad (55)$$

The generating function is the formal-power-series solution of

$$A(z)G(z)^2 + B(z)G(z) + C(z) = 0, \quad G(0) = \lambda. \quad (56)$$

Equivalently, put

$$D(z) = B(z)^2 - 4A(z)C(z)$$

and choose the formal square root $R(z)$ by

$$R(z)^2 = D(z), \quad R(0) = -\lambda^2\eta. \quad (57)$$

Then

$$G(z) = \frac{-B(z) + R(z)}{2A(z)}. \quad (58)$$

This normalization, rather than a fixed printed choice of a plus or minus sign, is essential. Indeed,

$$2A(0)\lambda + B(0) = -\lambda^2\eta,$$

so (57) makes (58) select precisely the root with constant coefficient $g_0 = \lambda$. When the square-root routine chooses the opposite constant term, its sign must be reversed.

For $n \geq 1$, form the ordinary Hankel determinants

$$H_n = \det(g_{i+j})_{0 \leq i, j \leq n-1}. \quad (59)$$

We find now that

$$H_1 = \lambda, \quad H_2 = 1, \quad H_3 = 1, \quad H_4 = \eta, \quad (60)$$

and

$$H_{n+4}H_n = H_{n+3}H_{n+1} + \tau H_{n+2}^2. \quad (61)$$

Thus the three parameters have distinct roles: τ is the second coefficient of the Somos family $A(1, \tau)$, whereas λ and η select the initial Hankel determinants (H_1, H_2, H_3, H_4) .

The quotient expression (58) requires $A(0) = \lambda(\tau + \lambda - \eta) \neq 0$. The branch normalization is nondegenerate when $\lambda\eta \neq 0$, because

$$\left. \frac{\partial}{\partial X} (A(z)X^2 + B(z)X + C(z)) \right|_{z=0, X=\lambda} = -\lambda^2\eta.$$

These are the admissibility conditions used implicitly by the formal-series construction.

13 Symplectic Elliptic Dynamics of the Jacobi Coefficients

Assume that the Hankel determinants a_n satisfy $A(p_1, p_2)$, and let $u_n = \widehat{b}_n = a_{n-1}a_{n+1}/a_n^2$. By Section 10,

$$u_{n+2} = \frac{p_4 u_{n+1} - p_1}{u_{n+1}^2} - u_n, \quad (62)$$

which defines the birational map

$$T(x, y) = \left(y, \frac{p_4 y - p_1}{y^2} - x \right). \quad (63)$$

Define

$$I(x, y) = x^2 y^2 - p_4 x y + p_1(x + y) + p_2. \quad (64)$$

Proposition 11. *The polynomial I is invariant under T : $I(T(x, y)) = I(x, y)$. Consequently, every level set $C_K : I(x, y) = K$ is invariant.*

Proof. Set $z = (p_4 y - p_1)/y^2 - x$. Then $x + z = (p_4 y - p_1)/y^2$, and

$$I(y, z) - I(x, y) = (z - x)(y^2(z + x) - p_4 y + p_1) = 0.$$

□

Proposition 12. *The map T preserves the standard symplectic form $\omega = dx \wedge dy$. Equivalently, $\det DT(x, y) = 1$ where T is defined.*

Proof. Writing $T(x, y) = (y, z)$ gives

$$DT(x, y) = \begin{pmatrix} 0 & 1 \\ -1 & -p_4/y^2 + 2p_1/y^3 \end{pmatrix},$$

whose determinant is 1. □

Corollary 13. *With respect to the canonical Poisson bracket, T is a Poisson map.*

Proposition 14. *The inverse is*

$$T^{-1}(x, y) = \left(\frac{p_4 x - p_1}{x^2} - y, x \right). \quad (65)$$

Moreover, T is reversible: if $\rho(x, y) = (y, x)$, then $\rho \circ T \circ \rho = T^{-1}$.

13.1 The invariant elliptic pencil

For a fixed K , the equation of C_K is

$$x^2y^2 + (p_1 - p_4x)y + p_1x + p_2 - K = 0. \quad (66)$$

Introduce $z = 2x^2y + p_1 - p_4x$. Completing the square gives

$$z^2 = -4p_1x^3 + (p_4^2 - 4p_2 + 4K)x^2 - 2p_1p_4x + p_1^2. \quad (67)$$

Theorem 15. *If the cubic on the right has no repeated root, the smooth projective completion of C_K is a curve of genus one, birational to the displayed cubic model.*

13.2 Involutions and QRT structure

Define

$$\iota_s(x, y) = (y, x), \quad \iota_v(x, y) = \left(x, \frac{p_4x - p_1}{x^2} - y\right).$$

Then $T = \iota_v \circ \iota_s$. Consequently, T is a symmetric QRT-type map associated with the biquadratic pencil C_K , in the sense of Quispel, Roberts, and Thompson [8].

13.3 A commuting elliptic Hamiltonian flow

Since T preserves both the symplectic form and I , consider

$$\dot{x} = \frac{\partial I}{\partial y}, \quad \dot{y} = -\frac{\partial I}{\partial x}. \quad (68)$$

Explicitly,

$$\dot{x} = 2x^2y - p_4x + p_1, \quad \dot{y} = -2xy^2 + p_4y - p_1. \quad (69)$$

Under the substitution above, $\dot{x} = z$ and the flow is elliptic. Since T is symplectic and $I \circ T = I$, T commutes with the Hamiltonian flow wherever both sides are defined.

14 Zero-Diagonal Prefix Convergents and Recovery of the Jacobi Orbit

We now exploit the freedom in the diagonal Jacobi coefficients by making the specific choice

$$\hat{d}_n = 0 \quad (n \geq 0).$$

This is not claimed to describe every moment functional covered by the characterization theorem. It selects a particularly tractable and mathematically interesting representative for which the relation between the Somos orbit and the prefix convergents can be examined directly.

Put $z = x^2$ and abbreviate $b_n = \hat{b}_n$. Consider the finite prefix convergents

$$C_n(z) = \frac{1}{1 - \frac{b_1z}{1 - \frac{b_2z}{\ddots - \frac{b_nz}{1}}}}, \quad C_0(z) = 1. \quad (70)$$

Thus all the C_n are prefixes of one and the same formal continued fraction. Let $Q_n(z)$ denote the corresponding denominator continuants, normalized by

$$Q_{-1}(z) = Q_0(z) = 1, \quad Q_n(z) = Q_{n-1}(z) - b_nzQ_{n-2}(z). \quad (71)$$

In particular, $Q_n(0) = 1$ for every n .

Proposition 16 (Successive-convergent difference). *For $n \geq 1$,*

$$C_n(z) - C_{n-1}(z) = \frac{(b_1 b_2 \cdots b_n) z^n}{Q_n(z) Q_{n-1}(z)}. \quad (72)$$

Consequently, if

$$R_n(z) := \frac{C_n(z) - C_{n-1}(z)}{z(C_{n-1}(z) - C_{n-2}(z))} \quad (n \geq 2), \quad (73)$$

then the apparent singularity at $z = 0$ is removable and

$$R_n(z) = b_n \frac{Q_{n-2}(z)}{Q_n(z)}, \quad R_n(0) = b_n. \quad (74)$$

Proof. The standard continuant determinant identity, or a direct induction using (71), gives (72). Dividing its n th instance by z times its $(n-1)$ st instance gives

$$R_n(z) = \frac{b_1 \cdots b_n}{b_1 \cdots b_{n-1}} \frac{Q_{n-2}(z)}{Q_n(z)}.$$

Since every Q_j has constant term 1, evaluation at $z = 0$ proves the last claim. $\square$

This gives an exact answer to one part of the convergent-level question. The Jacobi coefficient is recoverable from three consecutive prefix convergents, not by their common value, but by the first normalized difference that has not yet stabilized.

Theorem 17 (Somos law for normalized convergent increments). *Assume that the shifted Hankel determinants satisfy the Somos-type relation, so that*

$$F(b_n, b_{n+1}) = 0, \quad F(X, Y) = X^2 Y^2 - p_4 X Y + p_1(X + Y) + p_2.$$

Then, for every relevant n ,

$$F(R_n(0), R_{n+1}(0)) = 0, \quad (75)$$

where R_n is defined solely from four neighboring prefix convergents by (73). Equivalently,

$$F\left(\frac{C_n - C_{n-1}}{z(C_{n-1} - C_{n-2})}, \frac{C_{n+1} - C_n}{z(C_n - C_{n-1})}\right)\Big|_{z=0} = 0, \quad (76)$$

where both quotients are interpreted by removal of their singularities at $z = 0$.

Proof. By (74), $R_n(0) = b_n$ and $R_{n+1}(0) = b_{n+1}$. Substitution into the Jacobi biquadratic gives the claim. $\square$

The unnormalized differences tend to zero to successively higher orders, whereas division by the newly appearing power of z retains the Jacobi coefficient.

A useful approximation follows without evaluating the removable singularity exactly. From (71),

$$R_n(z) = b_n \left(1 + (b_{n-1} + b_n)z + O(z^2)\right). \quad (77)$$

Hence

$$F(R_n(z), R_{n+1}(z)) = O(z). \quad (78)$$

A first corrected estimator is

$$B_n^{[1]}(z) := \frac{R_n(z)}{1 + z(R_{n-1}(z) + R_n(z))}, \quad (79)$$

for which

$$B_n^{[1]}(z) = b_n + O(z^2), \quad F(B_n^{[1]}(z), B_{n+1}^{[1]}(z)) = O(z^2). \quad (80)$$

Thus finite evaluations of neighboring convergents provide systematically improvable approximations to the Jacobi orbit. This is useful when the convergents are available numerically or symbolically but extraction of their first non-stabilized coefficient is inconvenient.

There is also a direct error-amplitude interpretation. Since each additional level first contributes at order z^n , define

$$E_n := [z^n](C_n(z) - C_{n-1}(z)). \quad (81)$$

Equation (72) yields

$$E_n = b_1 b_2 \cdots b_n, \quad b_n = \frac{E_n}{E_{n-1}}. \quad (82)$$

The Somos biquadratic is therefore equivalent to

$$\left(\frac{E_n}{E_{n-1}}\right)^2 \left(\frac{E_{n+1}}{E_n}\right)^2 - p_4 \frac{E_{n+1}}{E_{n-1}} + p_1 \left(\frac{E_n}{E_{n-1}} + \frac{E_{n+1}}{E_n}\right) + p_2 = 0. \quad (83)$$

This is a closed Somos-type law for the leading increments of successive prefix approximants.

Remark 18. The literal substitution $C_{n-1} = C_n = C_{n+1} = S$ is not the appropriate test for the zero-diagonal case, because it is performed before the first non-stabilized coefficient is resolved. Equations (76) and (83) show that the dynamics survives in the normalized confluence of the convergents. This does not yet give a scalar algebraic equation for $S(z)$ alone. It gives instead an exact hierarchy of constraints on the successive approximation layers by which the prefix convergents build $S(z)$. Any further scalar equation for S must be compatible with this refined limit rather than inferred from the unscaled diagonal specialization.

15 Zero-Diagonal Jacobi Matrices and a Possible Volterra Lift

The diagonal Jacobi coefficients are not constrained by the Somos identity. One may therefore make the canonical choice $\hat{d}_n = 0$, giving

$$\hat{P}_{n+1}(t) = t\hat{P}_n(t) - \hat{b}_n\hat{P}_{n-1}(t). \quad (84)$$

Then $\hat{P}_n(-t) = (-1)^n \hat{P}_n(t)$. After a choice of square roots $\alpha_n^2 = \hat{b}_n$, the associated symmetric Jacobi matrix is zero-diagonal and bipartite. Such matrices are natural spectral operators for the Kac–van Moerbeke, or Volterra, hierarchy; see van Moerbeke [9] and Michor and Teschl [10]. In one standard normalization,

$$\dot{u}_n = u_n(u_{n+1} - u_{n-1}). \quad (85)$$

The present discrete evolution is reversible, symplectic, and preserves a pencil of genus-one curves, but a direct identification with the first Volterra flow should not be made without an additional argument. The contraction

$$u_n = \hat{b}_n = c_{2n}c_{2n+1} \quad (86)$$

forgets the factorization into individual Stieltjes coefficients. Writing $c_{2n} = r_n$ and $c_{2n+1} = u_n/r_n$ exposes an additional multiplicative variable invisible in the projected Jacobi dynamics.

Conjecture 19. *There exists a lift of the birational Jacobi map $T(u_n, u_{n+1}) = (u_{n+1}, u_{n+2})$ to a birational transformation of the Stieltjes coefficients c_j such that:*

1. *the contraction $u_n = c_{2n}c_{2n+1}$ recovers T ;*
2. *the lifted transformation possesses a Lax or Darboux representation of Volterra or modified-Volterra type;*

3. the biquadratic first integral I is obtained from a spectral invariant of the lifted transformation.

The example A377264 is particularly suggestive in this context. A first step toward proving the conjecture would be to determine the translation point induced by T on the Weierstrass model. A second would be to lift this translation to the factorization variables and compare the resulting transformation with known Volterra and modified-Volterra Darboux maps.

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